



INSTITUTE OF ENGINEERING & TECHNOLOGY

MOHANLAL SUKHADIA UNIVERSITY, UDAIPUR

Name : Lavina Tamboli | Class: Btech-CSE IV sem) | Subject: DBMS Lab

Experiment-3

AIM: Write a SQL statements for implementing ALTER, UPDATE and DELETE.

Step-1: Write a Query to show table employee_lavina.

```
mysql> select * from employee_lavina;
```

employee_id	first_name	last_name	salary	department_id	hire
1	John	Smith	60000	1	2022-01-15
2	Jane	Doe	75000	1	2021-03-20
3	Peter	Jones	90000	2	2020-05-10
4	Mary	Williams	80000	2	2021-08-01
5	David	Brown	65000	3	2023-01-05
6	NULL	David	55000	3	2023-02-12

6 rows in set (0.03 sec)

Step-2: Write a Query to add a new column to the employee_lavina table suing ALTER.

```
mysql> ALTER table employee_lavina ADD email varchar(50);  
Query OK, 0 rows affected (0.07 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

Step-3: Write a Query to rename a column using ALTER.

```
mysql> ALTER table employee_lavina RENAME COLUMN hire to hire_date;  
Query OK, 0 rows affected (0.03 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

Step-4: Write a Query to modify the data type of an existing column using ALTER.

```
mysql> ALTER table employee_lavina MODIFY COLUMN hire_date date;  
Query OK, 0 rows affected (0.02 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

Step-5: Write a Query to verify the table structure after all ALTER operations.

```
mysql> desc employee_lavina;
```

Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
first_name	varchar(20)	YES		NULL	
last_name	varchar(20)	YES		NULL	
salary	int	YES		NULL	
department_id	int	YES	MUL	NULL	
hire_date	date	YES		NULL	
email	varchar(50)	YES		NULL	

7 rows in set (0.00 sec)

Step-6: Write a Query to drop a column from the table using ALTER.

```
mysql> ALTER table employee_lavina DROP email;  
Query OK, 0 rows affected (0.02 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```



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Step-7: Write a Query to update the salary of a specific employee_lavina using UPDATE.

```
mysql> update employee_lavina
-> set salary = 70000
-> where employee_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Step -8: Write a Query to verify the table after all update operations.

```
mysql> select * from employee_lavina;
+-----+-----+-----+-----+-----+-----+
| employee_id | first_name | last_name | salary | department_id | hire_date |
+-----+-----+-----+-----+-----+-----+
| 1 | John | Smith | 70000 | 1 | 2022-01-15 |
| 2 | Jane | Doe | 75000 | 1 | 2021-03-20 |
| 3 | Peter | Jones | 90000 | 2 | 2020-05-10 |
| 4 | Mary | Williams | 80000 | 2 | 2021-08-01 |
| 5 | David | Brown | 65000 | 3 | 2023-01-05 |
| 6 | NULL | David | 55000 | 3 | 2023-02-12 |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

Step-9: Write a Query to delete a specific employee_lavina record using DELETE.

```
mysql> DELETE from employee_lavina
-> where employee_id = 6;
Query OK, 1 row affected (0.01 sec)
```

Step-10: Write a Query to delete all the employees of a particular department using DELETE.

```
mysql> DELETE from employee_lavina
-> where employee_id =3;
Query OK, 1 row affected (0.01 sec)
```

Step-11: write a Query to verify the table after DELETE operations.

```
mysql> SELECT * from employee_lavina;
+-----+-----+-----+-----+-----+-----+
| employee_id | first_name | last_name | salary | department_id | hire_date |
+-----+-----+-----+-----+-----+-----+
| 1 | John | Smith | 70000 | 1 | 2022-01-15 |
| 2 | Jane | Doe | 75000 | 1 | 2021-03-20 |
| 4 | Mary | Williams | 80000 | 2 | 2021-08-01 |
| 5 | David | Brown | 65000 | 3 | 2023-01-05 |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```



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Experiment-4

AIM: Write the query for implementing the following functions: MAX(), MIN(), AVG() and COUNT().

Step-1: Write the Query to show table employee_lavina.

```
mysql> select * from employee_lavina;
+-----+-----+-----+-----+-----+-----+
| employee_id | first_name | last_name | salary | department_id | hire |
+-----+-----+-----+-----+-----+-----+
| 1 | John | Smith | 60000 | 1 | 2022-01-15 |
| 2 | Jane | Doe | 75000 | 1 | 2021-03-20 |
| 3 | Peter | Jones | 90000 | 2 | 2020-05-10 |
| 4 | Mary | Williams | 80000 | 2 | 2021-08-01 |
| 5 | David | Brown | 65000 | 3 | 2023-01-05 |
| 6 | NULL | David | 55000 | 3 | 2023-02-12 |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

Step-2: Write a Query to find the maximum salary using MAX().

```
mysql> select MAX(salary) as Maximum_Salary from employee_lavina;
+-----+
| Maximum_Salary |
+-----+
| 90000 |
+-----+
1 row in set (0.00 sec)
```

Step-3: Write a Query to find the minimum salary using MIN().

```
mysql> select MIN(salary) as minimum_Salary from employee_lavina;
+-----+
| minimum_Salary |
+-----+
| 55000 |
+-----+
1 row in set (0.01 sec)
```

Step-4: Write a Query to calculate the average salary using AVG().

```
mysql> select AVG(salary) as average_Salary from employee_lavina;
+-----+
| average_Salary |
+-----+
| 70833.3333 |
+-----+
1 row in set (0.01 sec)
```

Step-5: Write a Query to count the total number of employees using COUNT().

```
mysql> select count(*) as total_employees from employee_lavina;
+-----+
| total_employees |
+-----+
| 6 |
+-----+
1 row in set (0.01 sec)
```



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Step-6: Write a Query the find the MAX salary department-wise using MAX() with GROUP BY.

```
mysql> select D.department_name, MAX(E.salary) as max_Salary
-> from employee_lavina E, departments D
-> where E.department_id = D.department_id
-> Group by D.department_name;
```

department_name	max_Salary
Engineering	75000
Sales	90000
Marketing	65000

3 rows in set (0.01 sec)

Step-7: Write a Query to find the MIN salary department-wise using MIN() with GROUP BY.

```
mysql> select D.department_name, MIN(E.salary) as min_Salary
-> from employee_lavina E, departments D
-> where E.department_id = D.department_id
-> Group by D.department_name;
```

department_name	min_Salary
Engineering	60000
Sales	80000
Marketing	55000

3 rows in set (0.00 sec)

Step-8: Write a Query to find the AVG salary per department using AVG() with GROUP BY.

```
mysql> select D.department_name, AVG(E.salary) as Avg_Salary
-> from employee_lavina E, departments D
-> where E.department_id = D.department_id
-> Group by D.department_name;
```

department_name	Avg_Salary
Engineering	67500.0000
Sales	85000.0000
Marketing	60000.0000

3 rows in set (0.01 sec)

Step-9: Write a Query to count employees in each department using COUNT() with GROUP BY.

```
mysql> select D.department_name, COUNT(*) as total_employees
-> from employee_lavina E, departments D
-> where E.department_id = D.department_id
-> Group by D.department_name;
```

department_name	total_employees
Engineering	2
Sales	2
Marketing	2

3 rows in set (0.00 sec)